

Exercise-4

Conditional Statements

a) Objective: Understand if and else Conditional statements in C.

```
#include<stdio.h>

int main()
{
    int num;
    scanf("%d",&num);
    if (num > 0)
    {
        printf("Positive number");
    }
    else
    {
        printf("Negative number");
    }

    return 0;
}
```

O/P

8

Positive number

b) Roots of a Quadratic Equation.

```
#include<stdio.h>

#include<math.h> // for sqrt() function

int main()
{
    float a,b,c,r1,r2,d;

    printf("Enter a,b,c Values:");
    scanf("%f%f%f",&a,&b,&c);
    d=b*b-4*a*c;
    if(d==0)
    {
        printf("roots are real & Equal \n");
        r1=-b/(2*a);
        r2=-b/(2*a);
        printf("root1=%f\n",r1);
        printf("root2=%f\n",r2);
    }
    else if(d>0)
    {
        printf("roots are real & different \n");
        r1=(-b+sqrt(d))/(2*a);
        r2=(-b-sqrt(d))/(2*a);
        printf("root1=%f\n",r1);
        printf("root2=%f\n",r2);
    }
}
```

```

    }
    else
    {
        printf("roots are imaginary\n");
    }
    return 0;
}

```

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Enter a,b,c Values:10 20 30

roots are imaginary

c) Generate electricity bill.

```

#include<stdio.h>

int main()
{
    int custno;
    float units,charges;
    printf("Enter Customer no:");
    scanf("%d",&custno);
    printf("Enter no.of units consumed:");
    scanf("%f",&units);
    if(units>=0 && units<=200)
    {
        charges=0.5*units;
    }
}

```

```

else if(units>=201 && units<=400)
{
    charges=100+(units-200)*0.65;
}
else if(units>=401 && units<=600)
{
    charges=230+(units-400)*0.80;
}
else
{
    charges=390+(units-600)*1.5;
}
printf("Amount to be paid is %f",charges);
return 0;
}

```

O/P

Enter Customer no:123

Enter no.of units consumed:100

Amount to be paid is 50.000000

d) Simulate a calculator using switch case.

//Arithmetic operators using switch case

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
int a,b,c;

printf("Enter a,b Values:");

scanf("%d%d",&a,&b);

printf("\n1.Addition\n2.Subtraction\n3.Multiplication\n4.Division\n5.Modulo Division");

printf("\n\n\t Enter your choice:");

scanf("%d",&c);

switch(c)
{
    case 1:
        printf("The Addition of %d and %d is:%d",a,b,a+b);
        break;
    case 2:
        printf("The Subtraction of %d and %d is:%d",a,b,a-b);
        break;
    case 3:
        printf("The Multiplication of %d and %d is:%d",a,b,a*b);
        break;
    case 4:
        printf("The division of %d and %d is:%d",a,b,a/b);
        break;
    case 5:
        printf("The Modulodivision of %d and %d
is:%d",a,b,a%b);
```

```
        break;
        default:
        printf("\n enter data properly");
        break;

    }
    return 0;
}
```

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Enter a,b Values:20 80

1.Addition

2.Subtraction

3.Multiplication

4.Division

5.Modulo Division

Enter your choice:3

The Multiplication of 20 and 80 is:1600

e) Find the given year is a leap year or not.

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
    int y;
```

```
scanf("%d",&y);  
if (((y % 4 == 0) && (y % 100 != 0)) || (y%400 == 0))  
    printf("It is a leap year");  
else  
    printf("It is not a leap year");  
return 0;  
}
```

O/P

2024

It is a leap year